"Fail fast. Recover smart. Stay resilient."

MODULE 32

Resilience and Fault Tolerance Patterns

Learn how to build resilient, fault-tolerant microservices with Resilience4j that stay reliable even when failures happen.

TEKsystems Global Services

WORKFORCE DEVELOPMENT

TEKsystems Global Services

WORKFORCE DEVELOPMENT

MODULE 32 OVERVIEW

Protect Spring Boot services from downstream failures using Resilience4j's Retry, Circuit Breaker, Time Limiter, and Fallback patterns.

Distributed systems fail constantly -- this module gives that reality a production-grade answer: detect failure fast, retry only what's safe, break the circuit before cascading failure, and always return a truthful, degraded response.

DURATION & LAB

1 Hour Theory

Failure modes, Retry with backoff, Circuit Breaker states, Time Limiter, and Fallback strategies.

2 Hours Lab

Resilience4j for CRM Outbound Calls: protect Account Profile reads and prove behavior with WireMock.

Scenario

CRM pages must degrade honestly when the Account API is down -- never lie about writes.

MODULE ARC

Detect Failures Fast

Understand why distributed calls fail and how a hanging thread pool cascades into an outage.

Master Resilience4j

Retry with backoff, Circuit Breaker states, Time Limiter, and truthful Fallback responses.

Build the Lab

Protect the CRM's Account Profile calls end to end and prove it with WireMock-driven tests.

KEY TAKEAWAY Total time: 3 hours (1 hour theory + 2 hours hands-on lab). By the end, you will design microservices that fail fast, recover smart, and stay resilient.

WHY DISTRIBUTED SYSTEMS FAIL

In a distributed environment, failures are not just possible -- they are inevitable.

THE REALITY OF DISTRIBUTED SYSTEMS

Unlike a single-server application, distributed systems consist of many independent components that run across networks and infrastructure beyond our direct control. They communicate over network calls, share data asynchronously, and depend on external services -- any of these dependencies can fail at any time.

COMMON REASONS WHY DISTRIBUTED SYSTEMS FAIL

Reason	Description
Network Issues	Latency, packet loss, DNS failures, or intermittent connectivity break service communication.
Component Failures	Services, databases, or instances can crash, become unresponsive, or run out of resources.
Timeouts	Slow responses or hanging calls cause timeouts and cascading failures across services.
External Dependency Failures	Third-party APIs, external services, or cloud resources may become unavailable or rate-limited.
Data Inconsistency	Partial updates, replication delays, or split-brain scenarios can lead to inconsistent data.
Overload Conditions	Traffic spikes or insufficient capacity can overwhelm services and degrade performance.
Configuration & Human Errors	Wrong configurations, bad deployments, or manual mistakes can introduce outages.
Infrastructure Problems	Disk failures, zone outages, or power issues can impact multiple services simultaneously.

KEY TAKEAWAY Failures are normal in distributed systems. The goal is not to prevent failures, but to detect them early, contain the impact, and recover quickly.

Why Distributed Systems Fail

1. Network Failures

- Packets can be lost, delayed, duplicated, or reordered
- Network partitions isolate parts of the system
- Timeouts and retries can cause cascading failures

2. Partial Failures

- Some nodes fail while others keep running
- Hard to detect and handle correctly
- Leads to inconsistent system state

3. Unreliable Clocks

- Clocks drift between nodes
- Events may appear out of order
- Breaks time-based logic and coordination

4. Concurrent Operations

- Multiple operations happen at the same time
- Causes race conditions and conflicts
- Hard to reason about system behavior

5. Data Inconsistency

- Replication and eventual consistency cause stale data
- Different nodes may have different views
- Leads to incorrect decisions and user confusion

6. Overload & Resource Limits

- High traffic or large loads exhaust resources
- Leads to timeouts, errors, and degraded performance
- Can trigger cascading failures

7. Unreliable Dependencies

- External services may be slow or unavailable
- Failures in dependencies impact the whole system
- Hard to isolate and recover

8. Configuration Errors

- Wrong configs cause outages or data loss
- Small changes can have big impact
- Hard to detect and rollback

9. Human Errors

- Mistakes in code, operations, or manual processes
- Unsafe deployments or bad data
- Hard to eliminate completely

10. Complexity

- Many moving parts and interactions
- Hard to test all scenarios
- Unknown unknowns cause surprises in production

MODULE LEARNING OBJECTIVES

By the end of this module, you will be able to design and implement resilient, fault-tolerant microservices that remain reliable under failure.

MODULE FORMAT

Lecture + Lab | **Theory: 1 Hour** | **Lab: 2 Hours** | **Total: 3 Hours**

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO:

#	Objective	Details
01	Understand System Failures	Explain why distributed systems fail and identify common failure scenarios.
02	Apply Retry Patterns	Implement retry mechanisms using fixed retry and exponential backoff strategies.
03	Implement Circuit Breakers	Design and configure circuit breakers and understand the Closed, Open, and Half-Open states.
04	Use Timeouts and Fallbacks	Configure timeouts and implement fallback strategies to ensure graceful degradation.
05	Leverage Resilience4j	Use Resilience4j to implement retries, circuit breakers, and other resilience patterns in Spring Boot.
06	Build Highly Available Systems	Apply enterprise resilience principles and best practices to build highly available, fault-tolerant microservices.

KEY TAKEAWAY Ultimate goal: design and build microservices that can withstand failures, recover quickly, and continue delivering a great experience to your users.

UNDERSTANDING SYSTEM FAILURES

Failures are a natural part of distributed systems. Understanding how and why they occur is the first step to building resilient applications.

TYPES OF SYSTEM FAILURES

Type	What It Means (with Example)
Transient Failures	Temporary issues that resolve on their own. Example: a brief network glitch or momentary service unavailability.
Partial Failures	Only part of the system is affected. Example: one instance is down while others succeed.
Intermittent Failures	Failures that come and go, making them hard to reproduce and diagnose. Example: a flaky third-party API.
Persistent Failures	Long-lasting issues requiring manual intervention. Example: a database outage, misconfiguration, or resource exhaustion.
Cascading Failures	A failure in one component triggers failures in others, amplifying the impact. Example: service timeout -> retries -> overload -> outage.

WHY DO SYSTEMS FAIL?

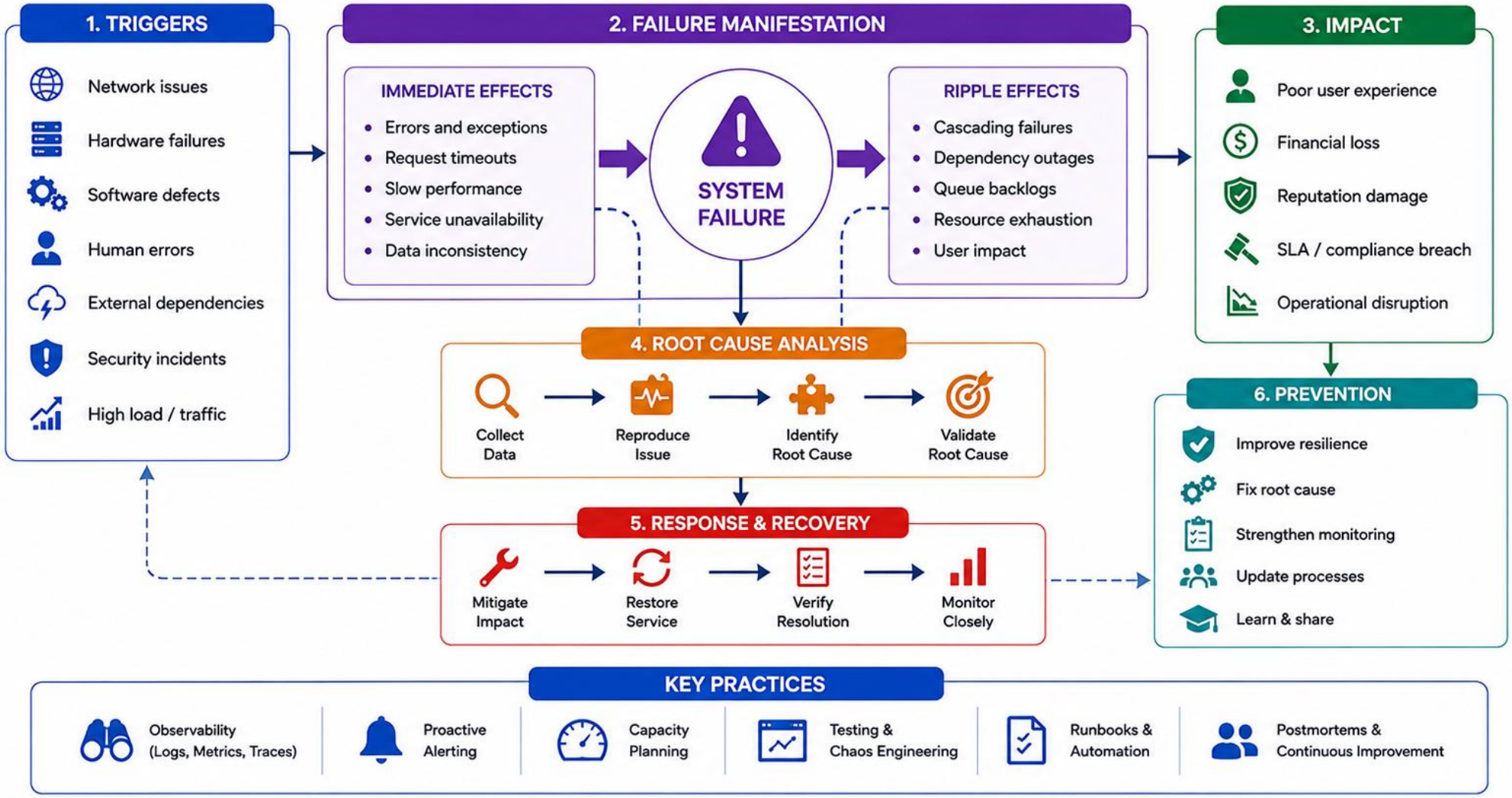
- Network -- latency, packet loss, DNS issues.
- Infrastructure -- disk, zone, power outages.
- Application -- bugs, resource leaks, bad deploys.
- Data -- inconsistency, corruption, replication lag.

IMPACT OF UNHANDLED FAILURES

- Service downtime and loss of availability.
- Poor user experience and lost trust.
- Data inconsistency or loss.
- Business and financial impact.

KEY TAKEAWAY Failures are inevitable. Resilience is a choice. Understand the types, sources, and impact of failures so you can design systems that detect, isolate, and recover quickly.

Understanding System Failures



COMMON FAILURE SCENARIOS

Failures can occur at different layers of a distributed system. Knowing the common scenarios helps us design the right resilience strategies.

Network Failures

Connectivity problems prevent services from communicating. E.g. high latency, packet loss, DNS resolution failure.

Service Unavailability

A service or instance is down or not reachable. E.g. service crash, instance terminated, health check failures.

Timeouts

A service takes too long to respond, causing the request to time out. E.g. slow queries, long external API calls.

Resource Exhaustion

A service runs out of essential resources. E.g. CPU saturation, memory leaks, disk full, connection pool exhausted.

Dependency Failures

An external dependency (service, API, database) fails or becomes slow. E.g. third-party API down, cache failure.

Data Inconsistency

Data state becomes inconsistent due to partial updates or concurrent operations. E.g. replication lag, stale reads.

Configuration Errors

Incorrect or missing configuration leads to component failures. E.g. wrong endpoint, invalid credentials.

Traffic Spikes

A sudden increase in traffic overwhelms the system. E.g. flash sales, viral promotions, improper rate limiting.

KEY TAKEAWAY Failures are inevitable in distributed systems. By understanding these common scenarios, we can design systems that detect, isolate, and recover quickly.

Common Failure Scenarios

1 Network Failure

- Packets are lost or delayed
- Connections time out
- Services become unreachable
- Can cause cascading failures

2 Server / Node Failure

- Hardware crashes or becomes unresponsive
- Services on the node stop
- Load shifts to remaining nodes
- May reduce system capacity

3 Service Failure

- Application or service crashes
- Errors increase, responses fail
- Dependencies may fail
- User requests are impacted

4 Database Failure

- Database becomes unavailable
- Queries fail or time out
- Data may be temporarily inaccessible
- Apps relying on DB are affected

5 Disk / Storage Failure

- Disk crashes or becomes full
- Data may be lost or corrupted
- Services cannot read / write
- Can lead to system downtime

6 Overload / High Traffic

- Too many requests or load spikes
- Resources (CPU, memory) exhausted
- Responses slow down or fail
- May degrade entire system

7 Dependency Failure

- External service or API is down
- Calls fail or return errors
- Features depending on it break
- Can trigger cascading failures

8 Data Inconsistency

- Data differs across replicas
- Updates not propagated
- Leads to incorrect results
- Hard to detect and fix

9 Configuration Error

- Wrong or missing configuration
- Services fail to start or connect
- Behavior is unexpected
- Hard to troubleshoot

10 Security Incident

- Unauthorized access or attack
- Data breach or service disruption
- Systems may be compromised
- Requires immediate response

DESIGNING FOR FAILURE (1 OF 2)

In distributed systems, failures are inevitable. The goal is not to prevent failures, but to design systems that can anticipate, withstand, and recover from them.

CORE PRINCIPLES

- **Expect Failure —** assume that any component can fail at any time.
- **Isolate Failure —** limit the blast radius so failures don't spread to the whole system.
- **Recover Quickly —** enable automatic recovery and graceful handling of failures.
- **Degrade Gracefully —** provide limited functionality instead of complete failure.
- **Observe Continuously —** monitor, detect, and gain visibility to act before users are impacted.

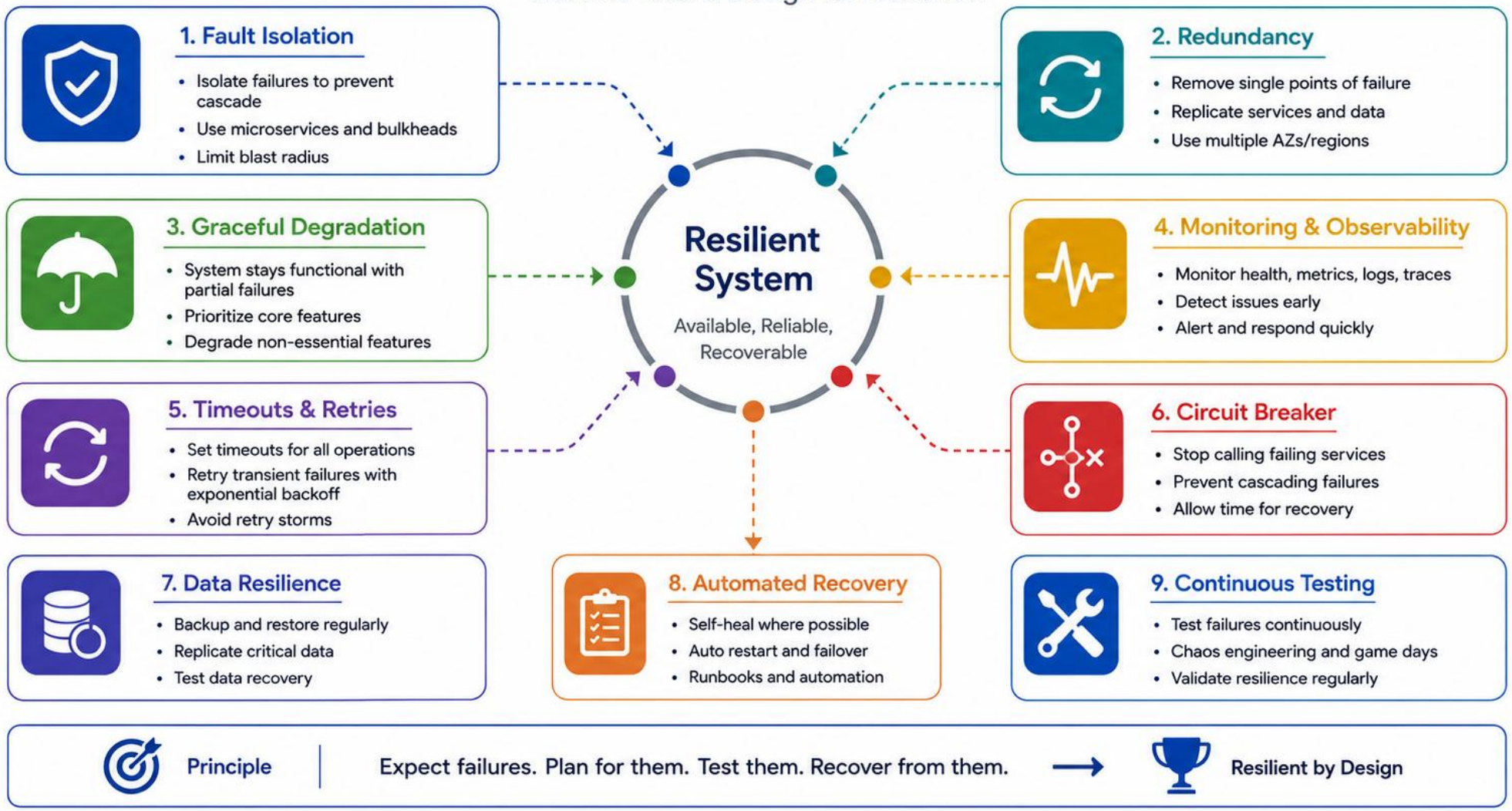
KEY DESIGN PRACTICES

Practice	What It Means
Redundancy	Remove single points of failure with replicas and multiple zones.
Loose Coupling	Design services with clear boundaries and minimal dependencies.
Async Communication	Use messaging to decouple services and absorb spikes.
Retries with Control	Retry transient failures with backoff and jitter to avoid overload.
Circuit Breakers	Stop calling failing services temporarily to allow recovery.
Timeouts	Set time limits for calls to prevent threads waiting indefinitely.
Fallbacks	Provide alternative responses or default behavior when needed.
Idempotency	Design operations to be safe to retry without duplicate effects.
Health Checks	Continuously verify the health of services and dependencies.

KEY TAKEAWAY Resilient systems are designed with failure in mind. Anticipate it. Isolate it. Recover from it. Keep delivering value.

Designing for Failure

Assume failure. Design for resilience.



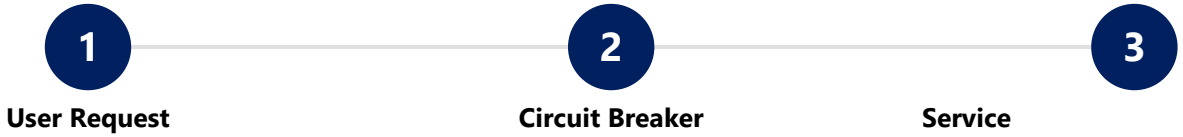
DESIGNING FOR FAILURE (2 OF 2)

A resilient design mindset, paired with a concrete example of these principles working together in a real request flow.

THE RESILIENT DESIGN MINDSET

- Build for resilience, not just for the happy path.
- Test for failure as much as you test for success.
- Embrace chaos in non-production environments.
- Continuously improve based on real-world incidents.

REAL-WORLD EXAMPLE: CIRCUIT BREAKER + FALLBACK



On success, the circuit breaker forwards the request and the service responds normally. On failure or timeout, the circuit breaker returns a Fallback Response instead of waiting or crashing.

WHY IT WORKS

When the service fails, the circuit breaker opens and a fallback response is returned to the user, ensuring a better experience than a hang or a crash.

KEY TAKEAWAY Resilient systems are designed with failure in mind: anticipate, isolate, recover, and keep delivering value -- even in a degraded state.

TRY IT YOURSELF — EXERCISE 1: WHY RESILIENCE

Reinforces: why an unprotected Account Profile call threatens the whole CRM request -- an analysis exercise.

THE SCENARIO

Customer detail for CUS-1001 (Amina Khan) calls the Account Profile service. The dependency hangs for 30 seconds -- and nothing in the CRM's code today bounds that wait, retries it, or protects the rest of the system while it happens.

STEPS (IN notes/lab32-resilience.md)

Step	What To Do
1 -- Hang Effects	List three user-visible or thread-pool effects of the 30-second hang (e.g. the customer detail page spins forever, the request thread is blocked and unavailable for other customers, health checks may start failing).
2 -- Name The Patterns	Write the four Resilience4j ideas this module teaches: Retry, Circuit Breaker, Time Limiter, Fallback.
3 -- Know The Limits	Write one sentence: resilience wraps calls -- it does not fix a permanently wrong URL or a broken configuration.
4 -- Save Your Work	Save the scenario analysis and pattern list under notes/lab32-resilience.md.

KEY TAKEAWAY TRY IT NOW — Complete Exercise 1 before continuing: exercises/exercise-01-why-resilience.md (workspace: examples/module-32-exercises).

RETRY PATTERN OVERVIEW

Retry patterns automatically re-attempt a failed operation that is likely to succeed when the failure is transient.

WHAT IS A RETRY PATTERN?

A retry pattern makes multiple attempts to complete an operation that has failed due to a temporary issue, such as network glitches, timeouts, or service unavailability.

Key idea: keep trying with control and intelligence, not blindly and indefinitely.

WHEN TO USE RETRY

- Transient failures (timeouts, network issues).
- Intermittent service unavailability.
- Rate limiting (HTTP 429).
- Optimistic concurrency conflicts.
- Short-lived infrastructure glitches.

Do NOT retry permanent failures: validation errors, 4xx client errors (except 429), invalid requests.

BENEFITS

- **Improved Reliability —** handles temporary issues automatically.
- **Better UX —** requests succeed without user intervention.
- **Resilience —** recover from short-lived failures gracefully.
- **Reduced Manual Effort —** less need for human monitoring and re-tries.

KEY CONSIDERATIONS

Retry only what is safe

Limit the number of retries

Add delay between retries (backoff)

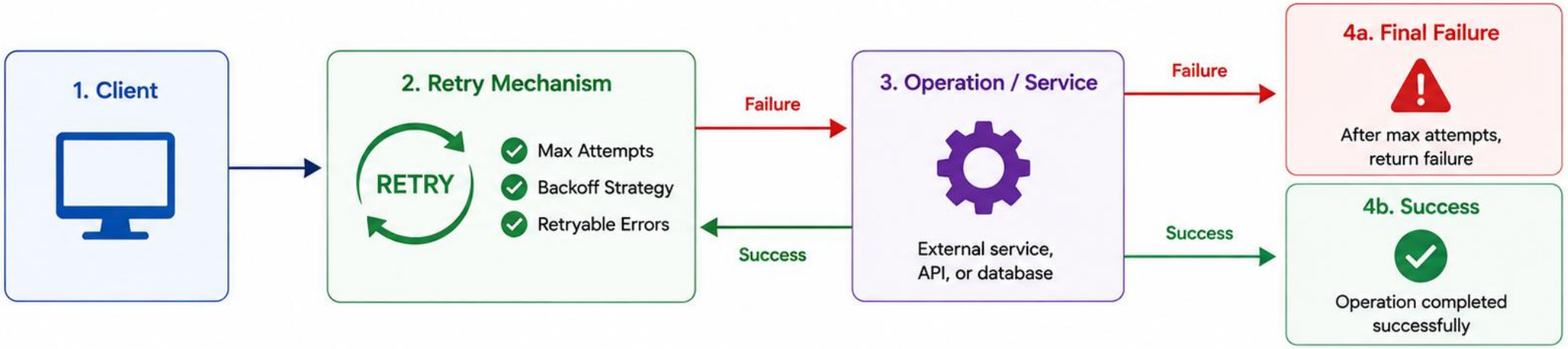
Add randomness (jitter)

Monitor and tune

KEY TAKEAWAY Use retry patterns wisely to handle transient failures and improve reliability, but always combine them with other resilience patterns like timeouts, circuit breakers, and fallbacks.

Retry Pattern Overview

Automatically retry a failed operation to improve reliability.



How It Works

1. Client makes a request
2. Retry mechanism calls the operation
3. If it fails, retry based on configured policy
4. If it succeeds, return result to client
5. If max attempts reached, return failure

Backoff Strategies

Fixed Backoff
Wait the same duration between retries

Exponential Backoff
Wait time increases exponentially

Exponential Backoff with Jitter
Adds random variation to reduce thundering herd

Retryable vs Non-Retryable Errors

Retryable Errors
- Network timeouts
- Temporary service unavailable (5xx)
- Rate limiting (429)

Non-Retryable Errors
- Bad request (4xx client errors)
- Authentication/authorization failures
- Validation errors

Benefits

- Improves reliability
- Handles transient failures
- Better user experience
- Reduces manual intervention

Best Practices

- Set appropriate max attempts
- Use exponential backoff with jitter
- Retry only idempotent operations
- Log each attempt
- Monitor retry success and failure rates

Key Takeaway: Retry Pattern helps handle temporary failures gracefully by retrying with a smart strategy.

RETRY STRATEGIES: FIXED VS EXPONENTIAL BACKOFF

Two common retry timing strategies -- a constant delay between attempts, or a delay that grows with each attempt.

FIXED RETRY -- CONSTANT DELAY

Retries a failed operation a fixed number of times with the SAME delay between each attempt. Simple, predictable, and good for brief transient glitches.

Attempt	Result	Next Action
1 (t=0s)	Fails	Wait 1s, retry
2 (t=1s)	Fails	Wait 1s, retry
3 (t=2s)	Fails	Wait 1s, retry
4 (t=3s)	Succeeds	Return result

maxRetries = 3, delay = 1s. If attempt 4 also failed, the caller would give up and return the error.

EXPONENTIAL BACKOFF -- INCREASING DELAY

Delay = baseDelay x 2^(attempt-1). Each retry waits longer than the last, reducing load and giving the system more time to recover. Add jitter to avoid a thundering herd.

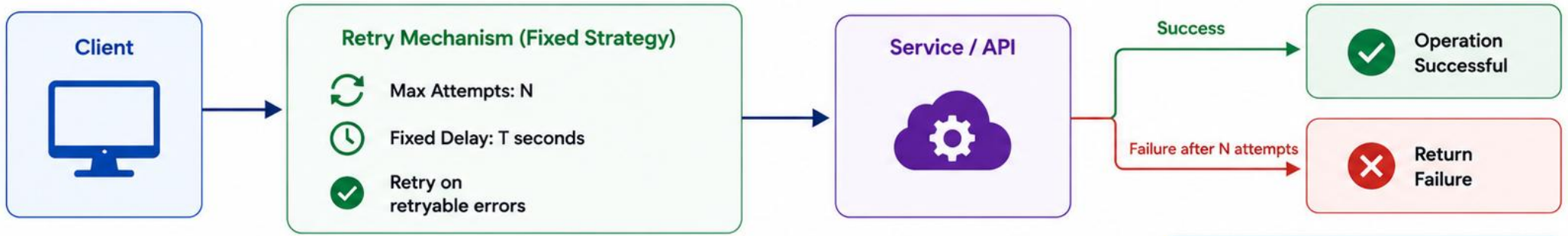
Attempt	Result	Next Wait
1 (t=0s)	Fails	Wait 1s (2^0)
2 (t=1s)	Fails	Wait 2s (2^1)
3 (t=3s)	Fails	Wait 4s (2^2)
4 (t=7s)	Fails	Wait 8s (2^3)
5 (t=15s)	Succeeds	Return result

baseDelay = 1s, multiplier = 2, maxAttempts = 5. Total wait before the last attempt = 1+2+4+8 = 15 seconds.

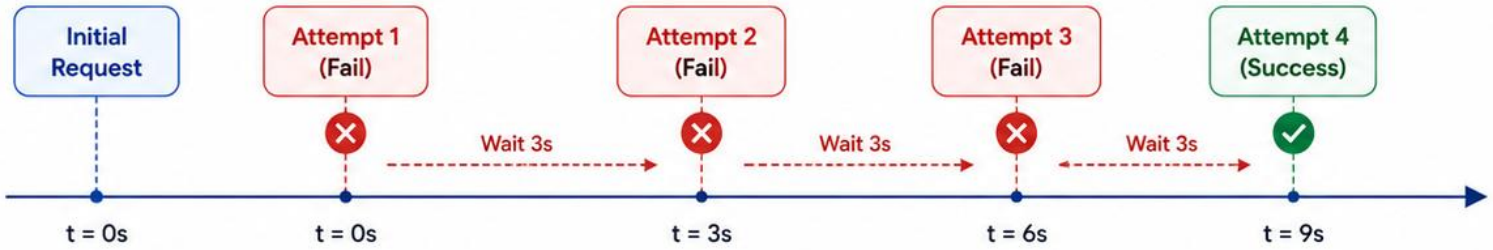
KEY TAKEAWAY Fixed retry is simple and predictable for short-lived glitches; exponential backoff (with jitter) scales the delay to reduce load and improve recovery odds under heavier or longer outages.

Fixed Retry Strategy

Retry a failed operation a fixed number of times with a constant delay between attempts.



How It Works (Example: Max Attempts = 4, Fixed Delay = 3 seconds)



Configuration Example

- maxAttempts: 4
- fixedDelay: 3000 ms (3 seconds)
- retryOn: [TimeoutException, Http 5xx, IOException]

Benefits

- Simple and easy to implement
- Predictable retry intervals
- Useful for stable, short-lived issues
- Reduces temporary failures

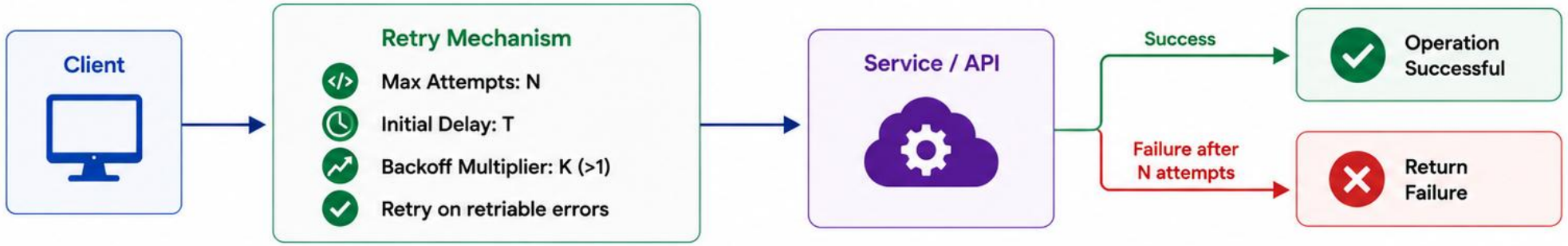
Use Cases

- Network glitches
- Temporary service unavailability
- Short transient outages
- External API timeouts

Key Takeaway: Fixed retry uses a constant delay between attempts.
It's simple, predictable, and effective for handling short-term transient failures.

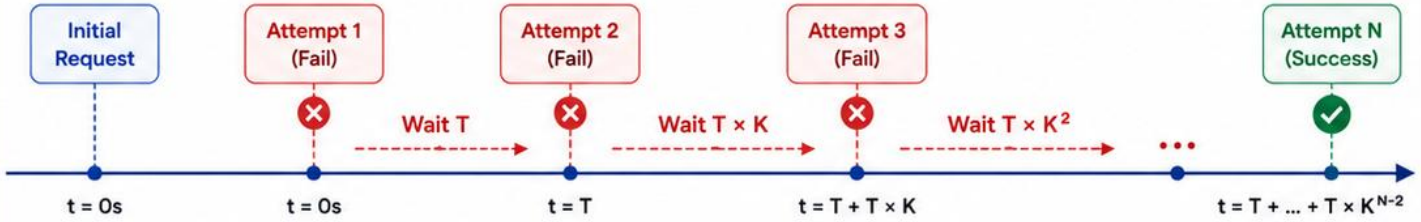
Exponential Backoff Strategy

Increase the wait time exponentially between retry attempts to reduce load and improve success rate.



How It Works (Example)

Formula: $Delay_n = T \times K^{(n-1)}$



Example Configuration

- maxAttempts: 5
- initialDelay: 1 s
- multiplier: 2.0
- maxDelay: 30 s (optional)

Attempt (n)	Delay Formula	Delay (seconds)	Cumulative Time (seconds)
1	$T \times 2^0$	1	0
2	$T \times 2^1$	2	1
3	$T \times 2^2$	4	3
4	$T \times 2^3$	8	7
5	$T \times 2^4$	16	15

Benefits

- Reduces load on failing services
- Prevents retry storms
- Increases success probability over time
- Adaptive and resilient

Use Cases

- Transient service outages
- Network timeouts
- Rate limited APIs (HTTP 429)
- Temporary infrastructure issues

Key Takeaway: Exponential backoff gives failing systems time to recover while minimizing unnecessary load.

WHAT IS A CIRCUIT BREAKER?

The Circuit Breaker pattern prevents a failing service from causing cascading failures by stopping requests to it for a period of time, allowing it to recover.

WHAT IS A CIRCUIT BREAKER?

A circuit breaker monitors calls to a service. When failures exceed a threshold, it 'opens the circuit' and blocks further calls for a configured time. After the timeout, it allows a few test calls to check if the service has recovered.

REAL-WORLD ANALOGY: ELECTRICAL CIRCUIT BREAKER

Goal: fail fast, reduce latency, and protect your system and its dependencies.

KEY BENEFITS

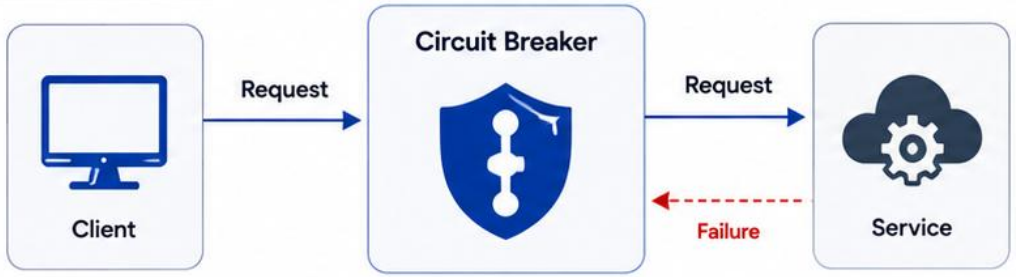
- Prevents Cascading Failures
- Improves System Resilience
- Reduces Latency & Waste
- Better User Experience
- Configurable & Observable

KEY TAKEAWAY A circuit breaker acts like a smart gatekeeper: it blocks traffic to a failing service, gives it time to heal, and only lets traffic through again when it's healthy. Remember: it's not about avoiding failures, it's about managing their impact.

What is a Circuit Breaker?

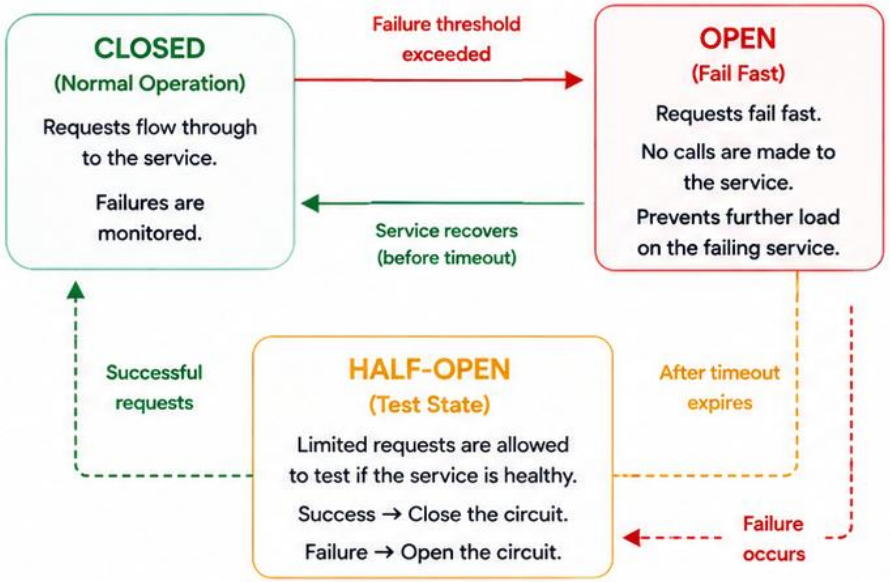
A circuit breaker prevents repeated failures and gives a failing service time to recover.

How It Works



- Prevents cascading failures
- Allows time for recovery
- Improves system resilience

Circuit Breaker States



Key Configuration Parameters

Failure Threshold	Number or percentage of failures that triggers opening the circuit.
Wait Duration (Open State)	Time the circuit stays open before trying again.
Half-Open Requests	Number of requests allowed in half-open state.
Success Threshold	Number of successful requests in half-open state to close the circuit.
Record Exceptions	Which exceptions count as failures.

Benefits

- Protects system from cascading failures
- Reduces load on failing services
- Provides time for recovery
- Improves overall system stability and resilience

In Short: The circuit breaker monitors failures and stops calling a failing service. It resumes calls after a timeout to check if the service has recovered.

CIRCUIT BREAKER STATES

A Circuit Breaker moves between three states -- Closed, Open, and Half-Open -- based on failures and timeouts, to protect your system and allow recovery.

1. **CLOSED**
   Healthy -- requests flow
2. **OPEN**
   Tripped -- requests blocked
3. **HALF-OPEN**
   Testing -- limited trial calls
4. **CLOSED (again)**
   Recovered -- normal flow

Closed -> Open when the failure threshold is exceeded within the sliding window. Open -> Half-Open after the wait duration elapses. Half-Open -> Closed if test calls succeed; Half-Open -> Open again if any test call fails.

TYPICAL CONFIGURATION

- **Failure Rate Threshold: e.g. 50%**
- **Minimum Number of Calls: e.g. 10**
- **Sliding Window: last 30s**
- **Wait Duration in Open: e.g. 30s**
- **Permitted Calls in Half-Open: e.g. 5**

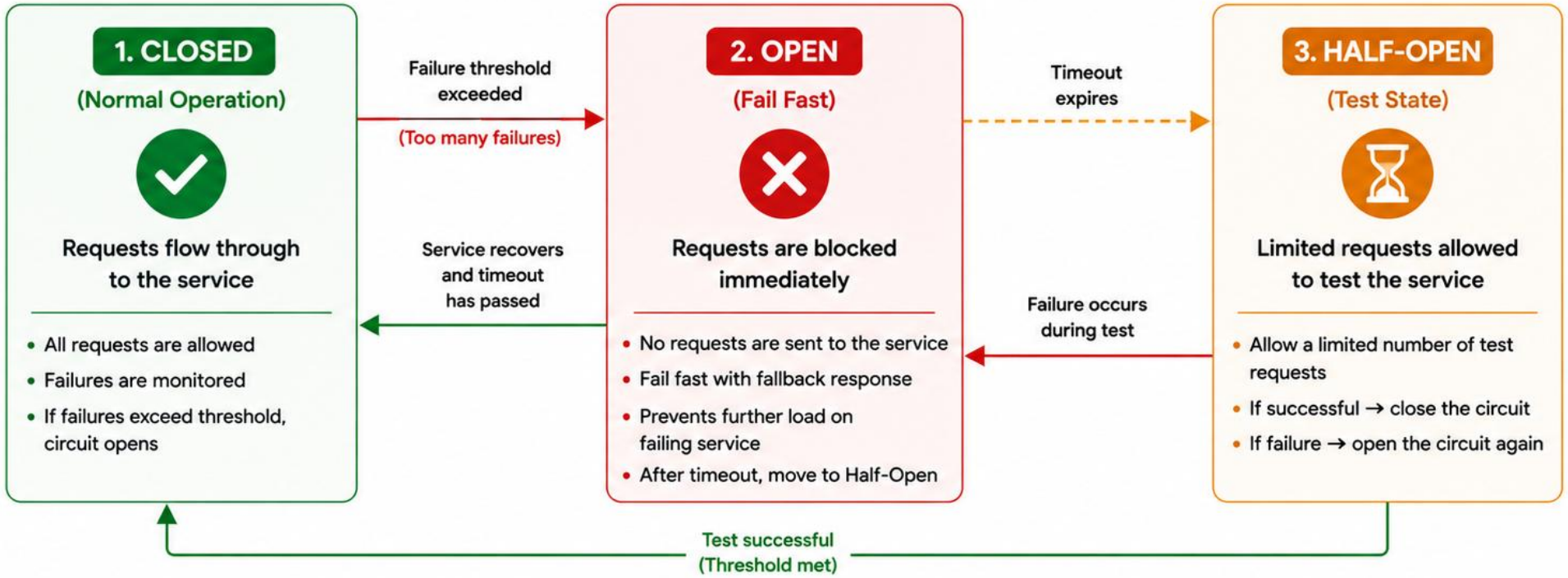
EXAMPLE SCENARIO: PAYMENT SERVICE IS DOWN

Circuit opens after 10 failures in 30s (threshold hit) -> all further calls fail fast for 30s -> after 30s, 5 test calls are allowed -> if they succeed the circuit closes; if they fail it opens again for another 30s.

KEY TAKEAWAY Circuit breaker states work together to detect failures early, stop the bleeding, and then cautiously restore traffic. Fail fast, stay resilient, recover smart.

Circuit Breaker States

Three states to protect your system and allow recovery



Goal

Prevent cascading failures and allow recovery

Benefits

- Protects downstream services
- Reduces system load
- Improves resilience

Key Configurations

- Failure Threshold
- Wait Duration (Open State)
- Half-Open Max Requests
- Success Threshold

Flow Summary

Closed → Open → Half-Open → Closed (Repeat as needed)

CIRCUIT BREAKER STATE DETAILS

Closed, Open, and Half-Open each have distinct behavior, purpose, and a trigger for moving to the next state.

State	Behavior	Purpose	Moves To...
CLOSED (Healthy)	All requests pass through and are forwarded to the service. Successes and failures are both counted; successful calls do NOT reset the failure count.	Allow traffic while the service is believed healthy, while continuously watching for trouble.	OPEN, if failures exceed the threshold within the sliding window.
OPEN (Tripped)	All incoming requests fail immediately without ever calling the service -- they fail fast.	Prevent overload and give the failing service time to recover without added load.	HALF-OPEN, once the configured wait duration has elapsed.
HALF-OPEN (Testing)	Only a limited number of test calls are allowed through; any additional requests during this window are rejected.	Safely probe the dependency to verify recovery without risking full traffic.	CLOSED if all test calls succeed; back to OPEN (with a fresh wait duration) if any test call fails.

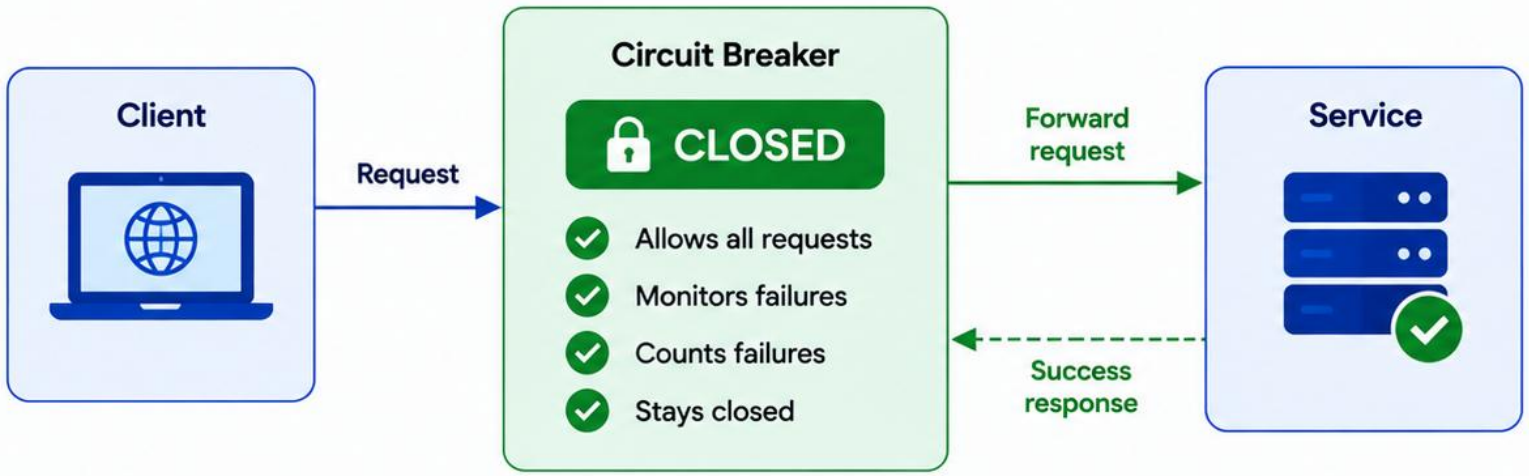
WORKED EXAMPLE: PAYMENT SERVICE

Payment Service is up: the circuit stays Closed while occasional failures are tracked below the threshold. Ten failures occur within 30 seconds, crossing the threshold -- the circuit Opens, and every request fails fast for 30 seconds. After 30 seconds, the circuit moves to Half-Open and allows 5 test requests: if they all succeed, the circuit Closes and normal traffic resumes; if any fails, it reopens for another 30 seconds.

KEY TAKEAWAY Closed is the default healthy state; Open is pure protection; Half-Open is a careful, limited probe. Succeed -> close and heal. Fail -> open again and protect.

Closed State

Normal operation – all requests are allowed and failures are monitored



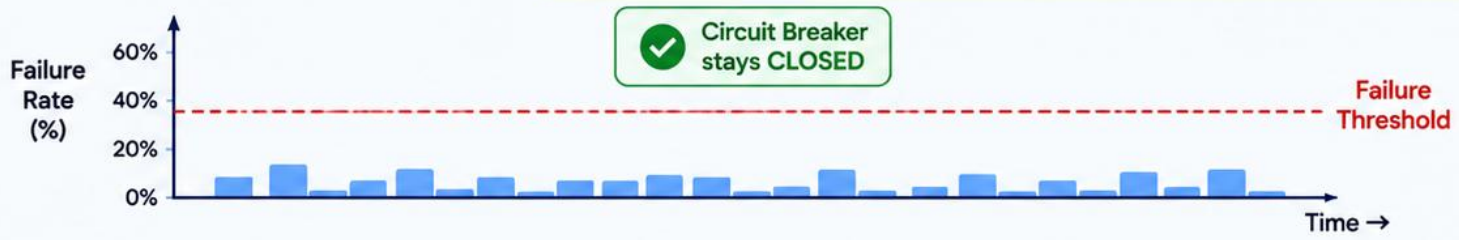
When is it Closed?

- Initial state
- Failure rate is below threshold
- Service is healthy
- After recovery from Half-Open

Key Characteristics

- Normal operation mode
- Zero latency added
- Continuously monitors service health
- Full traffic allowed

Failure Monitoring in Closed State



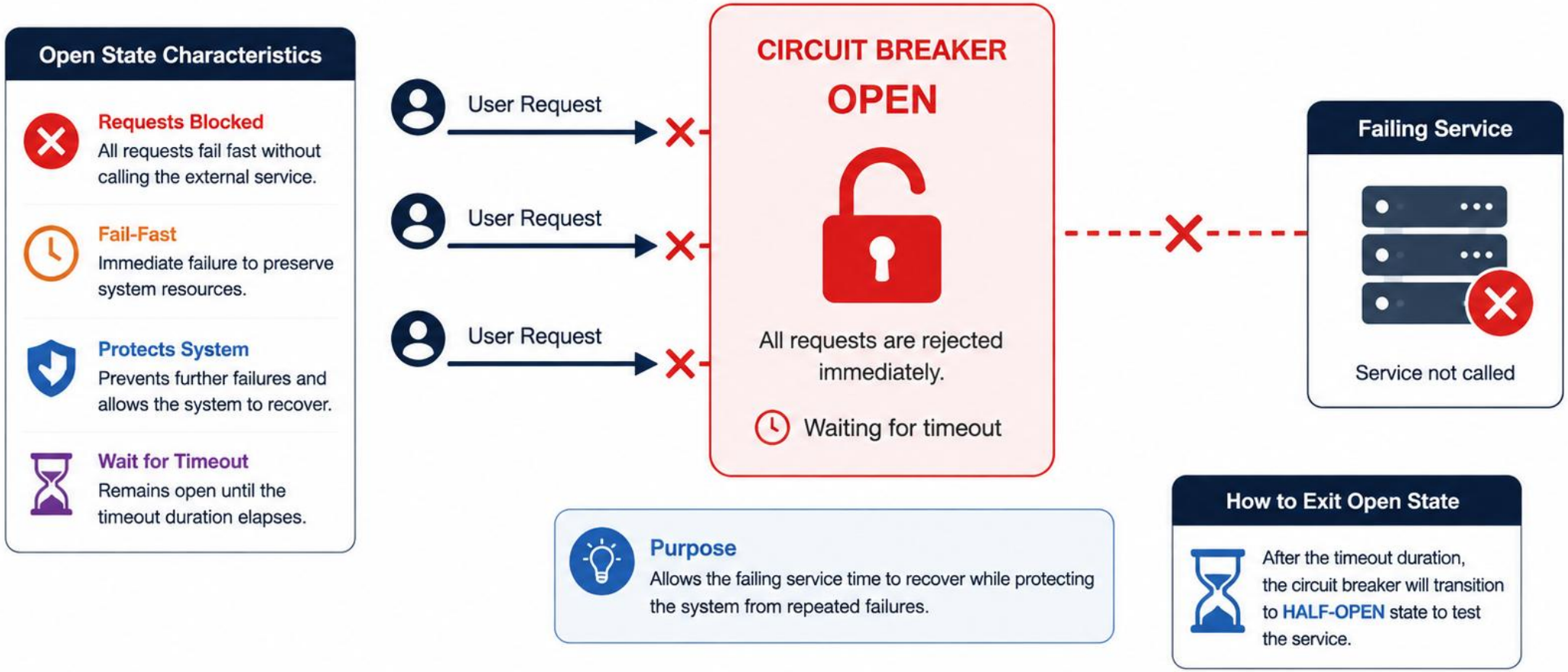
Transitions

If failure threshold is exceeded → Moves to OPEN state

Key Takeaway: The Closed state allows normal traffic while silently monitoring failures to detect service problems early.

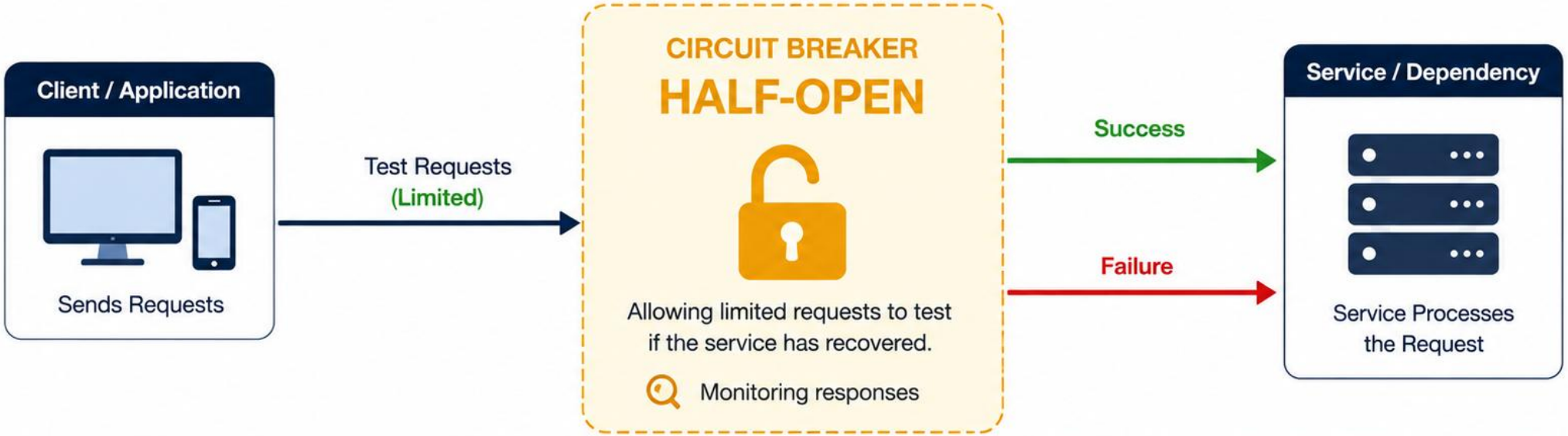
Open State

The circuit breaker is open. All requests are blocked immediately.
No calls are made to the failing service.



Half-Open State

The circuit breaker allows a limited number of test requests.
It monitors the responses to determine if the service has recovered.



If Successful

- Service is healthy.
- Circuit breaker closes.
- Normal operation resumes.

Transition to CLOSED State

If Failed

- Service is still unhealthy.
- Circuit breaker opens again.
- Requests are blocked.

Transition to OPEN State

Purpose

Half-open state prevents unnecessary load on a failing service while allowing recovery detection.

TRY IT YOURSELF — EXERCISE 2: CIRCUIT STATES

Reinforces: documenting Closed, Open, and Half-Open for the Account Profile breaker -- a documentation exercise.

STATE NOTES (IN notes/circuit-states.md)

State	What To Write
Closed	Normal calls flow to Account Profile; both successes and failures are counted.
Open	Calls fail fast or use the fallback; Account Profile is not hammered while it's unhealthy.
Half-Open	A small number of trial calls probe recovery; success -> Closed, failure -> Open again.

DRAW THE STATE DIAGRAM (BOXES + ARROWS)

```
[CLOSED]    --failure threshold exceeded-->       [OPEN]
[OPEN]      --wait duration elapses-------->   [HALF-OPEN]
[HALF-OPEN] --trial calls all succeed------->     [CLOSED]
[HALF-OPEN] --any trial call fails---------->        [OPEN]
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 2 before continuing: exercises/exercise-02-circuit-states.md (mention the fallback explicitly for the Open state).

TIMEOUT MANAGEMENT

Timeouts ensure that slow or unresponsive services don't block resources and degrade system performance. They define the maximum time to wait for a response.

WHY TIMEOUTS MATTER

- **Prevent Resource Exhaustion —** threads waiting too long can exhaust pools and degrade the entire system.
- **Improve Responsiveness —** fail fast when a service is slow, so the system stays responsive.
- **Enable Other Patterns —** timeouts are essential for retries, circuit breakers, and fallbacks to work.
- **Better UX —** users get a quick failure or fallback instead of a long, silent wait.

TIMEOUT TYPES

- **Connection Timeout —** maximum time to establish a connection. Example: 2 seconds.
- **Read Timeout —** maximum time to wait for a response after the request is sent. Example: 3 seconds.
- **Overall Timeout —** maximum total time for connect + send + wait. Example: 5 seconds.

TimeLimiter (application.yml)

```
resilience4j:
  timelimiter:
    instances:
      accountProfile:
        timeout-duration: 1500ms
        cancel-running-future: true
```

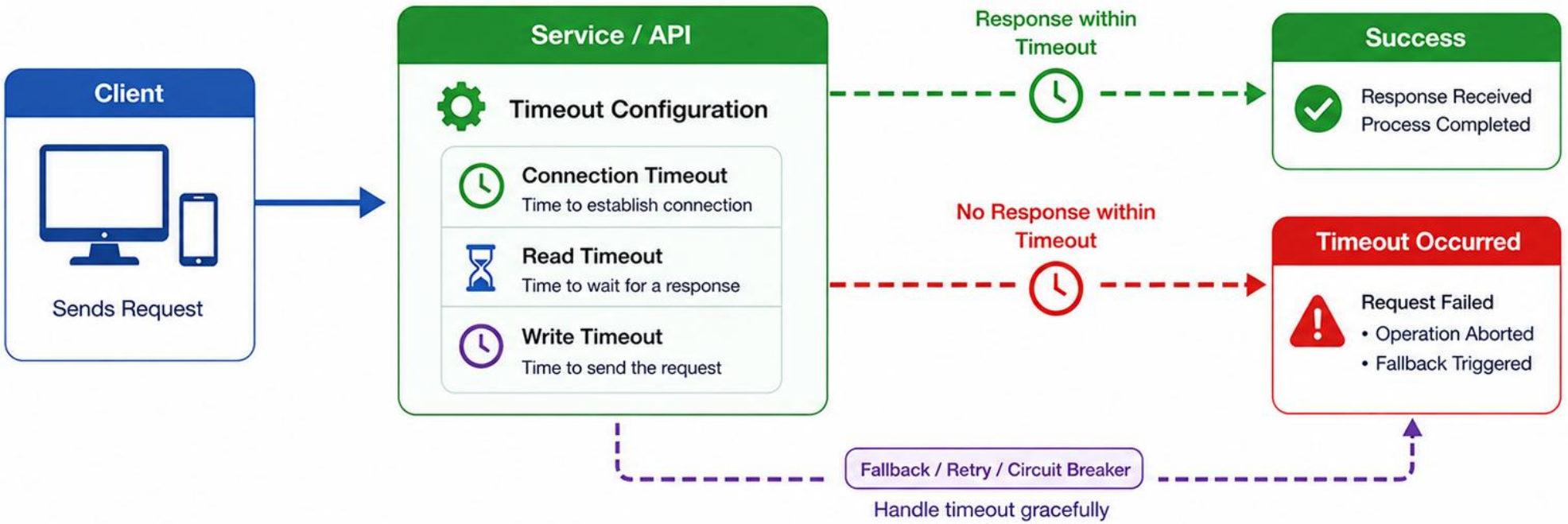
TIMEOUT IN ACTION

If the response is not received within the configured timeout, the call is aborted (cancelled) and control returns immediately -- to retry, fall back, or fail fast -- instead of hanging.

KEY TAKEAWAY Timeouts protect your system by ensuring no call waits forever. Set the right values, monitor them, and combine with retries and circuit breakers for true resilience.

Timeout Management

Proper timeout configuration prevents resource exhaustion and improves system reliability.



Timeout Best Practices

Set Appropriate Values
Configure timeouts based on service characteristics and performance.

Different Timeouts
Use different values for connection, read, and write operations.

Use with Retries
Combine timeouts with retry logic and exponential backoff.

Prevent Cascading Failures
Timeouts help stop slow services from affecting the entire system.

Monitor and Tune
Continuously monitor and adjust timeout values based on metrics.

FALLBACK STRATEGIES

Fallback strategies provide an alternative way to deliver value when the primary service is unavailable, slow, or returns an error.

WHY FALLBACKS MATTER

- **Business Continuity —** keeps the application functional even when external services fail.
- **Better UX —** users get a meaningful response instead of errors or long waits.
- **Reduced Load —** serve cached or default data quickly instead of waiting on a failed call.
- **Protects Downstream —** avoids cascading failures by providing a safe alternative.

COMMON FALLBACK STRATEGIES

- **Cached Response —** return data from cache when the primary service fails.
- **Default / Static —** return a predefined default value or message.
- **Stale Data —** return the last known good data, even if outdated.
- **Alternate Service —** use a different service or provider to fulfill the request.
- **Graceful Degradation —** return a partial response with limited functionality.

AccountService.java

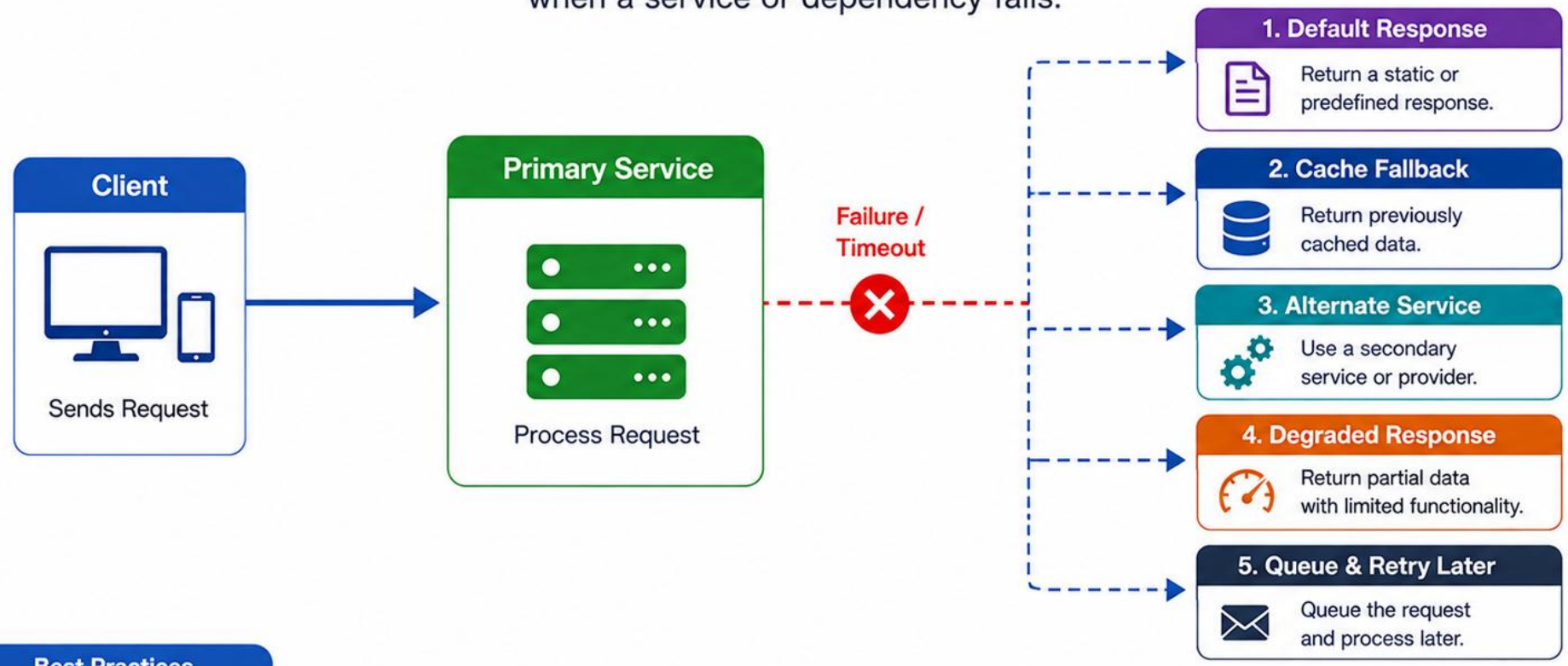
```
@CircuitBreaker(name = "accountProfile",
    fallbackMethod = "fallback")
public CompletableFuture<AccountSummary>
    find(String customerId) {
  return client.fetch(customerId);
}

private CompletableFuture<AccountSummary>
    fallback(String customerId, Throwable ex) {
  log.warn("account_profile_degraded");
  return CompletableFuture.completedFuture(
      AccountSummary.unavailable(customerId));
}
```

Fallbacks should be fast, reliable, and provide useful information -- even if it's degraded or partial.

KEY TAKEAWAY Fallbacks keep your application running and your users happy when things go wrong. Always provide a useful alternative, monitor it, and keep it simple.

Fallback Strategies

Fallback strategies ensure graceful degradation and a better user experience when a service or dependency fails.



Best Practices

Know Your System
Understand dependencies and failure points.

Choose the Right Fallback
Select fallback based on business needs.

Avoid Silent Failures
Log and monitor all fallback activations.

Monitor & Improve
Track fallback usage and refine strategies.

Test Your Fallbacks
Regularly test fallback scenarios.

GRACEFUL DEGRADATION

Graceful degradation keeps a service usable even when a dependency fails -- instead of breaking the whole request, return fallback or simplified data.

HOW IT WORKS

1. **Dependency Fails**
2. **Failure Detected**
3. **Fallback / Partial Data**
4. **Core App Continues**

When a non-critical dependency fails, show cached or partial data and keep the core application usable -- never show a broken interface or a raw error for something the user could otherwise still do.

WHEN TO USE

- External APIs unavailable
- Non-critical components fail
- Data is stale or incomplete
- Maintain core functionality

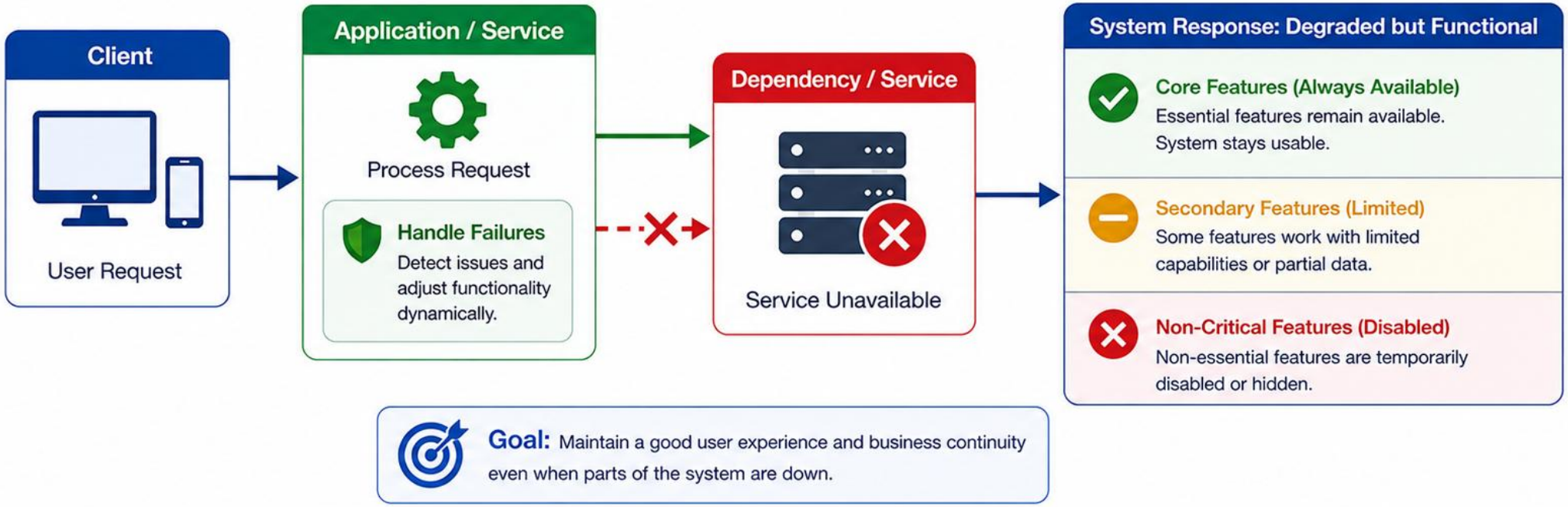
GRACEFUL DEGRADATION VS. TOTAL FAILURE

Graceful Degradation	Total Failure
Something works (partially).	Nothing works.
Users can still continue.	Users are blocked.
Better UX and retention.	High frustration.

KEY TAKEAWAY Graceful degradation ensures your application remains useful and reliable -- even when a downstream dependency goes wrong.

Graceful Degradation

The system continues to operate with reduced functionality
when a dependency or service is partially or completely unavailable.



Goal: Maintain a good user experience and business continuity even when parts of the system are down.

Best Practices

Prioritize Features
Identify and prioritize core features critical for users.

Set Fallbacks
Provide safe defaults, cached data, or alternate flows.

Communicate Clearly
Inform users about degraded mode and expected limitations.

Monitor & Adapt
Continuously monitor dependencies and adjust strategies.

Test Degradation
Regularly test failure scenarios to ensure graceful handling.

TRY IT YOURSELF — EXERCISE 3: FALLBACK CONTRACT

Reinforces: specifying an honest, minimal Account Profile response for Amina when the dependency fails -- a documentation exercise.

DEGRADED CONTRACT (IN notes/fallback-contract.md)

Decision	What To Write
Fields Kept	customerId, displayName (if cached), status = UNKNOWN.
Fields Dropped	balance, tier, lastLogin -- anything that could be stale or wrong is omitted, not guessed.
API Signal	Choose HTTP 200 with degraded=true, or HTTP 503 -- write one sentence justifying your choice.
User Message	Draft one UI string, e.g. "Account details temporarily limited."

WHY THIS MATTERS

A fallback that looks like a normal success response is worse than an honest error -- Lab 32's own rubric treats a fallback that implies a successful write as an honor violation, not just a bug.

KEY TAKEAWAY TRY IT NOW — Complete Exercise 3 before continuing: exercises/exercise-03-fallback-contract.md.

INTRODUCTION TO RESILIENCE4J (1 OF 2)

Resilience4j is a lightweight, fault-tolerance library for Java that helps you build resilient applications with easy-to-use, composable components.

KEY FEATURES

- **Lightweight —** minimal dependencies and low overhead.
- **Modular —** choose only the resilience components you need.
- **Functional Style —** designed for Java 8+, functional interfaces, and lambdas.
- **Micrometer Compatible —** built-in metrics integrate with Prometheus, Grafana, etc.
- **Cloud-Native Ready —** perfect for microservices, reactive, and distributed systems.

CORE COMPONENTS

Component	What It Does
Circuit Breaker	Detects failures; blocks calls to a failing service until it recovers.
Retry	Retries a failed operation a configurable number of times before giving up.
Time Limiter	Limits execution time of a call and cancels it if it exceeds the timeout.
Rate Limiter	Controls the rate of calls to a service and prevents overload.
Bulkhead	Limits concurrent calls to isolate resources and prevent cascading failures.

HIGH-LEVEL ARCHITECTURE

Resilience4j components (Circuit Breaker, Retry, Time Limiter, Rate Limiter, Bulkhead) wrap the call to the external service, and emit metrics, logs, and events. Components can be used individually or combined.

KEY TAKEAWAY Resilience4j gives you the building blocks to make your services resilient, scalable, and production-ready. Use the right component for the right problem, and combine them for maximum resilience.

Introduction to Resilience4j

Resilience4j is a lightweight, functional library for building resilient applications in Java.

What is Resilience4j?

Resilience4j provides a set of fault tolerance patterns to help your application handle failures gracefully and maintain stability in distributed systems.

RESILIENCE4J

Core Modules

Circuit Breaker

Prevents cascading failures by stopping calls to failing services and allows recovery detection.

Retry

Retries failed operations with configurable attempts and wait intervals.

Rate Limiter

Limits the number of calls to a service within a configured time window.

Bulkhead

Isolates resources by restricting concurrency and preventing resource exhaustion.

Time Limiter

Limits the execution time of calls and cancels long-running operations.

Key Benefits

- Improves application resilience and stability
- Prevents cascading failures
- Enhances user experience
- Lightweight and easy to integrate
- Functional programming friendly

How It Works

Why Resilience4j?

- 100% Java 8+ compatible
- Lightweight with no external dependencies
- Highly configurable and modular
- Works well with Spring Boot and other frameworks

INTRODUCTION TO RESILIENCE4J (2 OF 2)

Why teams choose Resilience4j, a concrete worked example, and how it fits into the wider observability ecosystem.

WHY USE RESILIENCE4J?

- Improve system availability and reliability.
- Handle failures gracefully instead of crashing.
- Prevent cascading failures in distributed systems.
- Improve user experience with fast, honest failure.
- Gain visibility into failures with metrics and events.
- Highly customizable and production-ready.

EXAMPLE USE CASE: ORDER SERVICE -> PAYMENT SERVICE

- **Circuit Breaker —** prevents calls when Payment Service is down.
- **Retry —** attempts the call again for transient failures.
- **Time Limiter —** ensures the call does not hang indefinitely.
- **Rate Limiter —** protects Payment Service from too many requests.
- **Bulkhead —** isolates Payment calls from other operations.

ECOSYSTEM INTEGRATION

Spring Boot **Prometheus** **Grafana** **OpenTelemetry** **Micrometer Metrics**

KEY TAKEAWAY Use the right Resilience4j component for the right problem, and combine them for maximum resilience -- Lab 32 combines Circuit Breaker, Retry, and Time Limiter on a single method.

TRY IT YOURSELF — EXERCISE 4: PATTERN MAP

Reinforces: assigning each Resilience4j component to a concrete Northstar Account Profile behavior.

PATTERN -> CRM USE (COPY INTO notes/pattern-map.md)

Pattern	CRM Use
Retry	Transient 503 from the Account Profile service.
TimeLimiter	Fail fast if the call exceeds the configured budget (N ms).
CircuitBreaker	Stop calling Account Profile once the failure rate is high.
Fallback	Return a cached or minimal profile for Amina (CUS-1001).

REMAINING STEPS

- **Ravi Row —** Add one example sentence for CUS-1002 (Ravi Singh) describing what the CRM returns while the circuit is open.
- **Decorator Order —** Propose a one-line order, e.g. TimeLimiter -> CircuitBreaker -> Retry -> call.
- **Boundary —** Note: do not apply a circuit breaker to a local in-memory map lookup -- only to real outbound calls.

KEY TAKEAWAY TRY IT NOW — Complete Exercise 4 before continuing: exercises/exercise-04-pattern-map.md.

RETRY WITH RESILIENCE4J

Automatically retry failed operations to improve reliability -- configurable via application.yml and applied with a single annotation.

RETRY CONFIGURATION (application.yml)

```yaml
resilience4j:
  retry:
    instances:
      accountProfile:
        max-attempts: 3
        wait-duration: 250ms
        enable-exponential-backoff: true
        exponential-backoff-multiplier: 2
        retry-exceptions:
          - java.io.IOException
          - c.n.crm.account.TemporaryAccountException
```

Map HTTP 503 to a TemporaryAccountException in the client. Never list business/validation errors as retryable.

RETRY + CIRCUIT BREAKER + TIME LIMITER (SPRING BOOT)

```java
@CircuitBreaker(name = "accountProfile",
    fallbackMethod = "fallback")
@Retry(name = "accountProfile")
@TimeLimiter(name = "accountProfile")
public CompletableFuture<AccountSummary> find(
    String customerId) {
  return CompletableFuture.supplyAsync(
      () -> client.fetch(customerId), executor);
}
```

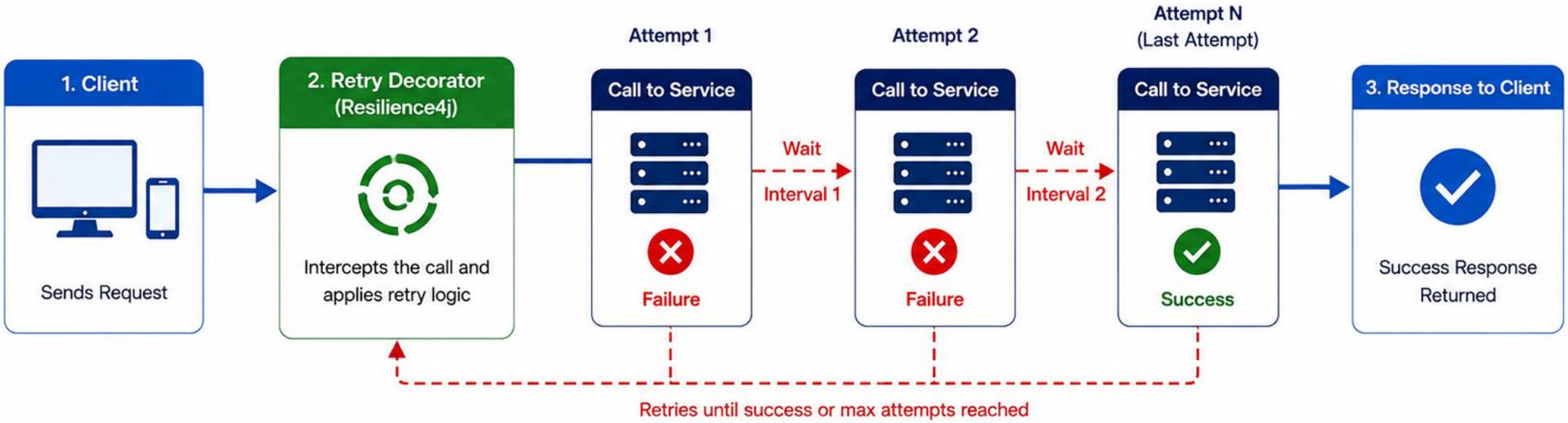
BEST PRACTICE

- **Retry only transient failures**
- **Never retry 4xx client errors**
- **Use exponential backoff**
- **Combine with Circuit Breaker + Timeout**

KEY TAKEAWAY Retry improves the success rate for transient failures, but only for idempotent, safe-to-repeat operations -- never blindly retry writes.

Retry with Resilience4j

Resilience4j Retry helps your application automatically retry failed operations with configurable attempts and wait intervals.



Retry Configuration (Example)

maxAttempts: 3
waitDuration: 1s
retryExceptions:
- IOExecption
- TimeoutException

- **waitDuration:** Time to wait between retries
- **maxAttempts:** Maximum number of retry attempts
- **retryExceptions:** Exceptions to retry on
- **ignoreExceptions:** Exceptions not to retry on

Benefits

- Improves reliability of remote calls
- Handles transient failures gracefully
- Configurable and easy to integrate
- Works well with Circuit Breaker and other patterns

CIRCUIT BREAKER WITH RESILIENCE4J

Resilience4j provides a lightweight Circuit Breaker that monitors calls and short-circuits requests when failures cross a configured threshold.

CIRCUIT BREAKER CONFIGURATION (application.yml)

```
resilience4j:
  circuitbreaker:
    instances:
      accountProfile:
        sliding-window-type: COUNT_BASED
        sliding-window-size: 10
        minimum-number-of-calls: 5
        failure-rate-threshold: 50
        wait-duration-in-open-state: 10s
        permitted-number-of-calls-in-half-open-state: 2
        automatic-transition-from-open-to-half-open-
          enabled: true
```

HOW IT WORKS WITH RESILIENCE4J

CLOSED: calls pass through, failures are recorded. OPEN: calls are rejected immediately (CallNotPermittedException); the fallback runs. HALF-OPEN: a limited number of trial requests are allowed through.

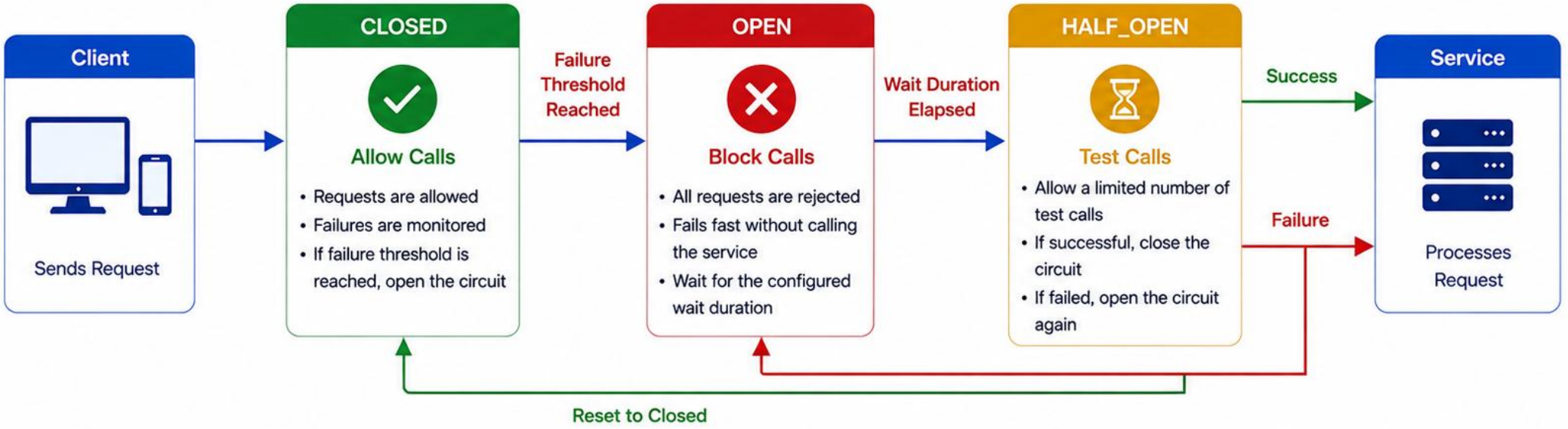
COMMON USE CASES

- External REST APIs
- Third-party services
- Database / cache calls
- Protecting critical microservices

KEY TAKEAWAY Resilience4j Circuit Breaker helps your system fail fast and recover gracefully. Configure thresholds from real-world metrics, and always pair it with a fallback.

Circuit Breaker with Resilience4j

Resilience4j Circuit Breaker prevents cascading failures by stopping calls to a failing service and allowing recovery when the service becomes available again.



Key Benefits

- Prevents cascading failures and instance overload
- Improves system resilience and availability
- Fails fast and conserves resources
- Automatically recovers when the service is healthy again
- Highly configurable and easy to integrate

Example Configuration

failureRateThreshold: 50
slidingWindowSize: 10
minimumNumberOfCalls: 5
waitDurationInOpenState: 10s
permittedNumberOfCallsInHalfOpenState: 3
slidingWindowType: COUNT_BASED

How it works: Monitor failures → Open circuit on threshold → Block calls → Wait → Test calls → Close on success or reopen on failure.

TRY IT YOURSELF — EXERCISE 5: FILL RESILIENCE TODOS (STARTER)

Reinforces: completing the Resilience4j annotation and config blanks for the CRM Account Profile client -- a fill-in-the-blank starter, not a complete solution.

STARTER -- FILL IN THE BLANKS (notes/lab32-todos.md)

```
@CircuitBreaker(name = "_____", fallbackMethod = "_____")
@Retry(name = "accountProfile")
@TimeLimiter(name = "accountProfile")
public CompletableFuture<AccountProfile> getProfile(String customerId) {
  return _____.fetch(customerId); // remote client
}
private CompletableFuture<AccountProfile> profileFallback(
    String customerId, Throwable t) {
  // TODO: return minimal profile for CUS-1001 / CUS-1002
  return CompletableFuture.completedFuture(_____);
}
```

STEPS (IN notes/lab32-todos.md)

Step	What To Do
1 -- Fill Annotations & Fallback	Suggested fills: accountProfile, profileFallback, accountClient, AccountProfile.minimal(customerId).
2 -- Config Blanks	Add YAML TODOs -- failureRateThreshold, waitDurationInOpenState, maxAttempts -- with example numbers you choose.
3 -- Correlation	Add a TODO comment: log lab-request-001 when the fallback fires.

KEY TAKEAWAY TRY IT NOW — Complete Exercise 5 (STARTER) before continuing: exercises/exercise-05-fill-resilience-todos.md. Fallback must match the return type plus one extra Throwable argument.

BUILDING HIGHLY AVAILABLE SYSTEMS (1 OF 2)

Highly Available (HA) systems are designed to minimize downtime and ensure continuous service delivery, even in the presence of failures.

WHY HIGH AVAILABILITY MATTERS

- **Better User Experience —** users expect services to always be available.
- **Business Continuity —** avoid revenue loss and protect brand reputation.
- **Fault Tolerance —** systems continue to operate even when components fail.
- **Scalability & Growth —** HA architectures support growth without compromising reliability.
- **Compliance & SLAs —** meet regulatory requirements and SLA commitments.

WHAT HIGH AVAILABILITY MEANS

HA is not just about uptime -- it's about minimizing downtime and ensuring fast, seamless recovery.

KEY PRINCIPLES OF HIGH AVAILABILITY

1. **Redundancy**
   No single points of failure
2. **Failover**
   Switch to healthy resources
3. **Monitoring**
   Continuous health checks
4. **Recovery**
   Automated recovery
5. **Scalability**
   Scale for demand

KEY TAKEAWAY Highly available systems eliminate single points of failure, detect issues early, and recover automatically -- keeping the service responsive for users.

Building Highly Available Systems

Highly available systems are designed to minimize downtime and ensure continuous service delivery even in the face of failures.

Core Principles

Redundancy
Eliminate single points of failure with redundant components.

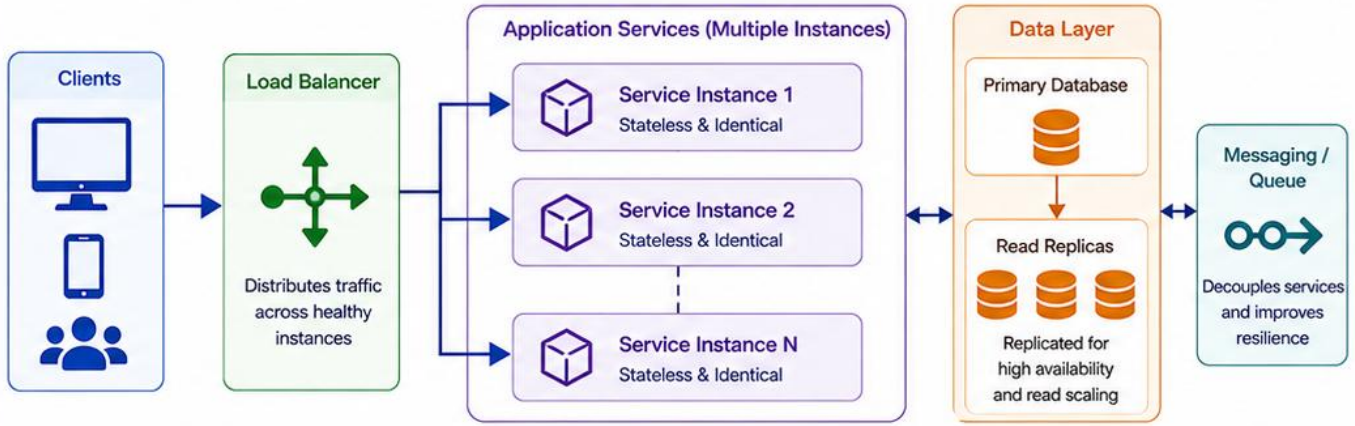
Fault Tolerance
Design systems to continue operating even when parts fail.

Failover
Automatically switch to backup resources on failure.

Monitoring
Continuously monitor health and performance to detect issues early.

Scalability
Scale horizontally to handle increasing load and ensure availability.

Highly Available Architecture



Health Checks
Continuously check the health of services and route traffic to healthy instances only.

Auto Recovery
Automatically restart failed instances and replace unhealthy components.

Multi-AZ / Region
Distribute resources across availability zones or regions for maximum availability.

Backups
Regular backups and point-in-time recovery to prevent data loss.

Observability
Centralized logging, metrics, and alerting for proactive issue resolution.

Benefits

- Minimizes downtime and service interruptions
- Improves system reliability and resilience
- Handles failures gracefully
- Enhances user experience and trust
- Supports business continuity and growth

Key Practices

Design for Failure
Assume failures will happen and design systems to handle them.

Eliminate Single Points
Remove single points of failure across all layers of the application.

Test Failures Regularly
Perform chaos engineering and failover testing to validate resilience.

Document & Automate
Document runbooks and automate recovery and operational tasks.

Monitor & Improve
Continuously monitor, analyze and improve system reliability over time.

BUILDING HIGHLY AVAILABLE SYSTEMS (2 OF 2)

HA architecture patterns, how Resilience4j's components support HA, and the metrics that prove it's actually working.

HIGH AVAILABILITY PATTERNS

Pattern	What It Means
Active-Active	Multiple active instances handle traffic. If one fails, others continue without interruption.
Active-Passive	Standby systems take over only when the active system fails.
Multi-Region	Deploy across regions for disaster tolerance and latency optimization.
Auto-Scaling	Dynamically scale resources based on demand to maintain availability.

RESILIENCE4J'S ROLE IN HA SYSTEMS

Component	HA Contribution
Circuit Breaker	Prevents cascading failures by stopping calls to unhealthy services.
Retry	Automatically retries transient failures to increase success rate.
Time Limiter	Ensures services respond within acceptable timeouts.
Rate Limiter	Protects services from overload by controlling request rate.
Bulkhead	Isolates resources and prevents impact on other services.

AVAILABILITY METRICS TO TRACK

Uptime % | **MTTR (Mean Time To Recover)** | **MTBF (Mean Time Between Failures)** | **Error Rate** | **Latency**

KEY TAKEAWAY Building highly available systems is about people, process, and technology working together. Design for failure, build with resilience, and deliver uninterrupted value.

RESILIENCE BEST PRACTICES

Resilient systems anticipate failures, absorb the impact, and recover quickly. Follow these best practices across design, isolation, observability, and recovery.

#	Practice	What It Means
1	Design for Failure	Assume components will fail; eliminate single points of failure; set timeouts and limits.
2	Isolate and Contain	Use Bulkhead and Circuit Breaker patterns to stop cascading failures from spreading.
3	Ensure Observability	Centralize logs with correlation IDs; collect key metrics; use distributed tracing and alerts.
4	Build for Recovery	Automate recovery with health checks; plan for failover; back up data; test disaster scenarios.
5	Manage Dependencies	Know, classify, and continuously monitor every external dependency your service relies on.
6	Optimize Performance	Right-size resources, use caching, load test regularly, and scale appropriately.
7	Practice Governance	Define SLAs/SLOs, document runbooks, and learn from every incident to improve continuously.
8	Combine Resilience4j Patterns	Circuit Breaker + Retry + Time Limiter + Rate Limiter + Bulkhead together give layered protection.

PRINCIPLES TO REMEMBER

Fail Fast **Resist Failure** **Recover Fast** **Learn & Improve**

KEY TAKEAWAY Resilience is not a single feature -- it's a mindset and a set of practices. Design for failure, isolate failures, observe everything, and automate recovery to build truly resilient systems.

Resilience Best Practices

Design, build, and operate systems that can withstand failures and recover quickly.

1 Design for Failure

Assume failures will happen. Design systems that can tolerate and recover from failures gracefully.

2 Eliminate Single Points of Failure

Remove single points of failure by adding redundancy across all layers.

3 Monitor Everything

Continuously monitor health, performance, and business metrics to detect issues early.

4 Automate Recovery

Automate failover, retries, scaling, and recovery processes to reduce manual intervention.

5 Use Resilience Patterns

Implement patterns like circuit breaker, retry, timeout, bulkhead, and fallback.

6 Keep It Simple

Avoid unnecessary complexity. Simple systems are easier to operate and more resilient.

7 Test for Resilience

Regularly perform failure testing, chaos engineering, and disaster recovery drills.

8 Graceful Degradation

Ensure the system continues to deliver partial functionality during failures.

9 Protect Dependencies

Isolate and limit the impact of failing dependencies using timeouts, bulkheads, and circuit breakers.

10 Learn and Improve

Continuously learn from incidents and improve systems, processes, and tooling.

Resilient systems are intentional, not accidental.

People
Build a culture of ownership and learning.

Process
Define clear processes for handling failures.

Technology
Use the right tools and patterns for resilience.

Practice
Continuously validate and improve resilience.

TRY IT YOURSELF — EXERCISE 6: LAB 32 READINESS (CAPSTONE)

Reinforces: confirming you can explain the patterns before wiring Resilience4j into the Lab 32 starter -- a pre-lab self-check.

READINESS CHECKLIST (IN notes/lab32-readiness.md)

Step	What To Do
1 -- Teach-Back	Explain the Circuit Breaker to a peer in four sentences or fewer -- write the sentences down.
2 -- Evidence Preview	Note the evidence you'll capture in the lab: logs showing the fallback firing for Amina, and a recorded breaker state change.
3 -- No Run Yet	Confirm this pre-lab did not require starting the Account Profile mock or Spring Boot.
4 -- Pass Mark	You pass this check if your pattern map (Exercise 2) and filled TODOs (Exercise 4) both exist in your notes.

READY FOR LAB 32?

Teach-back written | **Pattern map (Ex 2) done** | **TODOs (Ex 4) filled** | **Mock not started yet**

KEY TAKEAWAY TRY IT NOW — Complete Exercise 6 before continuing: exercises/exercise-06-lab32-readiness.md. This is the last checkpoint before Lab 32 -- Resilience4j for CRM Outbound Calls.

LAB OVERVIEW -- RESILIENCE AND FAULT TOLERANCE

Lab 32: Resilience4j for CRM Outbound Calls -- Northstar Account Profile

BUSINESS SCENARIO

- Agents opening Amina Khan (CUS-1001) see account summaries from a separate Account Profile service. When it flaps, the CRM must still show identity/status plus a clear "accounts temporarily unavailable" banner -- never a spinning tab or a lie that a balance updated.
- Leadership freezes the contract: outbound account reads are wrapped with Retry + CircuitBreaker + TimeLimiter; degraded reads return AccountSummary.unavailable; writes are never silently marked successful.
- Depends on Labs 25-31's CRM API -- prefer Lab 31 so Kafka and resilience coexist in the same project.

LEARNING OBJECTIVES

- Reproduce fast, slow, and failing responses with WireMock.
- Configure Resilience4j Retry for transient, safe GET requests.
- Configure a CircuitBreaker; observe CLOSED, OPEN, HALF_OPEN.
- Apply a TimeLimiter that enforces the CRM latency budget.
- Write explicit degraded fallbacks (AccountSummary.unavailable).
- Prevent unsafe retries and false-success fallbacks for writes.

WHAT YOU BUILD

- Copy lab31-crm to lab32-crm; add Resilience4j Boot3 + AOP + Actuator + WireMock; stub 503/slow/OK for CUS-1001; configure Retry/CircuitBreaker/TimeLimiter instance "accountProfile"; annotate AccountProfileService.find with @CircuitBreaker+@Retry+@TimeLimiter returning CompletableFuture; implement fallback -> AccountSummary.unavailable(customerId); expose Actuator health/metrics/events; write AccountProfileResilienceTest proving OPEN, timeout, and recovery.

KEY TAKEAWAY Fixtures CUS-1001 (Amina Khan) and CUS-1002 (Ravi Singh) anchor every example; correlation ID lab-request-001 ties requests and logs together. Format: Lecture + Lab -- Theory 1 Hour, Lab 2 Hours, Total 3 Hours.

LAB TASKS AND DELIVERABLES

Resilience4j for CRM Outbound Calls -- implementation steps, deliverables, and the evaluation rubric.

#	Implementation Step
1	Branch lab31-crm to lab32-crm; add resilience4j-spring-boot3 + spring-boot-starter-aop + actuator.
2	Create deterministic WireMock stubs: 503-then-recover, permanent 503, and a 3000ms slow response for CUS-1001.
3	Configure bounded Retry (accountProfile): max-attempts 3, wait-duration 250ms, exponential backoff x2.
4	Configure CircuitBreaker (accountProfile): count-based window of 10, min calls 5, 50% failure threshold, 10s wait in Open.
5	Configure TimeLimiter (accountProfile): 1500ms timeout, cancel-running-future true; call the client via CompletableFuture.
6	Compose @CircuitBreaker + @Retry + @TimeLimiter on AccountProfileService.find(); call through the Spring bean, never this.
7	Implement fallback() -> AccountSummary.unavailable(customerId); never mark a write as falsely successful.
8	Expose Actuator health/circuitbreakerevents/metrics; write AccountProfileResilienceTest proving OPEN, timeout, and recovery twice.

EXPECTED DELIVERABLES

- Resilience4j Retry + CircuitBreaker + TimeLimiter on account profile reads.
- AccountSummary.unavailable truthful fallback.
- WireMock stubs for 503 / slow / OK on CUS-1001.
- Actuator observation evidence.
- AccountProfileResilienceTest output (OPEN, timeout, recovery); no secrets committed.

RUBRIC (100 MARKS)

- Environment 10 · Core Impl 30 · Integration/Config 15 · Failure Handling 15 · Verification 10 · Security/Prod Awareness 10 · Docs 10

KEY TAKEAWAY A fallback that implies write success is an honor violation; a circuit breaker configured but never demonstrated OPEN is incomplete; sleep-only tests lose quality marks.

MODULE 32 SUMMARY

In this module, we learned how to design resilient systems, handle transient and persistent failures, and apply Resilience4j patterns to build highly available microservices.

KEY CONCEPTS COVERED

Detect Failures

Identify and handle failures gracefully; failures are inevitable, resilience is a choice.

Retry Smartly

Use fixed and exponential-backoff retry patterns for safe, idempotent, transient failures.

Break the Chain

Stop cascading failures with a Circuit Breaker's Closed / Open / Half-Open states.

Set Timeouts & Fallbacks

Prevent hanging calls with smart timeouts; degrade gracefully with truthful fallbacks.

Leverage Resilience4j

Configure Retry, CircuitBreaker, TimeLimiter, RateLimiter, and Bulkhead via annotations and YAML.

Stay Observable

Monitor health, metrics, and events so resilience patterns are visible, not silent.

MODULE FLOW RECAP

1. **Understand Failures**
2. **Retry + Backoff**
3. **Circuit Breaker**
4. **Timeouts + Fallbacks**
5. **Lab: CRM Outbound Calls**

KEY TAKEAWAY Resilience is not a feature -- it's a mindset and a practice. Design for failure, implement resilience patterns, observe continuously, and deliver value without interruption.

KNOWLEDGE CHECK (1 OF 2)

Test your understanding of resilience goals, retry, circuit breaker triggers, and rate limiting.

1. What is the primary goal of resilience in distributed systems?

A. Increase system complexity

B. Prevent all failures

C. Continue delivering value despite failures

D. Reduce hardware costs

2. Which pattern automatically retries a failed call with a backoff strategy?

A. Circuit Breaker

B. Retry

C. Rate Limiter

D. Bulkhead

3. When does a Circuit Breaker stop allowing calls to a service (open)?

A. When the service is slow

B. When the failure rate exceeds the threshold

C. When retry attempts reach the limit

D. When the system is under high load

4. Which pattern limits the number of requests sent in a given time window?

A. Time Limiter

B. Rate Limiter

C. Bulkhead

D. Fallback

KEY TAKEAWAY Resilience means continuing to deliver value despite failures; Retry re-attempts with backoff; a Circuit Breaker opens on failure-rate threshold; a Rate Limiter caps request volume per window.

KNOWLEDGE CHECK (2 OF 2)

Four more questions on Time Limiter, Bulkhead, the Half-Open state, and safe-to-retry failures.

5. What is the purpose of a Time Limiter?

A. To retry failed calls

B. To limit the number of concurrent calls

C. To restrict the rate of requests

D. To limit the maximum duration of a call

6. What does the Bulkhead pattern help to do?

A. Retry failed requests

B. Isolate resources and prevent failure propagation

C. Limit response time

D. Return fallback responses

7. In a Circuit Breaker, which state allows only a limited number of test calls to check whether a failing service has recovered?

A. Closed

B. Open

C. Half-Open

D. Disabled

8. Per Resilience4j best practice, which of the following should generally NOT be configured as retryable?

A. A network IOException

B. A 503 Service Unavailable response

C. A validation-driven 4xx client error

D. A socket timeout

KEY TAKEAWAY Time Limiter bounds call duration; Bulkhead isolates resources; Half-Open safely probes recovery with limited test calls; 4xx client/validation errors should never be retried.

"Fail fast. Recover smart. Stay resilient."

MODULE 32 COMPLETE

Resilience and Fault Tolerance Patterns

You can now design for failure, apply Retry, Circuit Breaker, Timeout, and Fallback patterns with Resilience4j, and build highly available microservices that keep delivering value.